

Complete Binary Tree

A **complete binary tree** is a binary tree where:

1. All levels are completely filled, except possibly the last level.
2. The last level is filled from left to right (no gaps in between).

👉 In other words: you can only leave empty spots at the very end of the last level.

Example (Complete):

Code >

Copy



- Levels 0 and 1 are fully filled.
- Last level has nodes 4, 5, 6 filled **from left to right**.
✅ This is a complete binary tree.

Non-Complete Binary Tree

A **non-complete binary tree** is any binary tree that **does not** satisfy the above rules.

That means:

- If a level before the last is not fully filled → ❌ not complete.
- If the last level has nodes but they are not filled left-to-right (i.e., a gap in the middle) → ❌ not complete.

Example (Not Complete):

Code >

Copy



\
5

- Node **2** has a right child but no left child.
- This breaks the “fill from left to right” rule.
✗ Not a complete binary tree.

✓ Quick Revision Notes

Python >

Copy

```
//& Title: Complete vs Non-Complete Binary Tree

//? Complete Binary Tree
/* All levels fully filled except possibly the last
/* Last level filled strictly from left to right
/* Example: Heap data structures use complete binary trees

//? Non-Complete Binary Tree
/* Any binary tree that breaks the above rules
/* Gaps in levels before the last OR gaps in the middle of the last level
/* Example: A node with a right child but no left child → not complete
```